

Eigenvalues and Eigenvectors

- ▶ $A \in \mathbb{R}^{n \times n}$. $\lambda \in \mathbb{R}$ is said to be an eigenvalue of A if $\exists x \neq 0$ such that $Ax = \lambda x$. x is an eigenvector associated with λ .
- ▶ For every $\lambda \in \mathbb{R}$, $x \in \mathbb{R}^n : x \neq 0$,
- ▶ $Ax = \lambda x \iff (A - \lambda I_n)x = 0 \implies p_A(\lambda) := \det(A - \lambda I_n) = 0$.
- ▶ spectrum of $A = \Lambda(A) := \{\lambda : p_A(\lambda) = 0\}$.
- ▶ λ is an eigenvalue of $A \iff \lambda \in \Lambda(A) \cap \mathbb{R}$
- ▶ x is an eigenvector of $A \iff x$ is a non-zero solution of $(A - \lambda I_n)x = 0$.
- ▶ E_λ =eigen-space associated with λ = Null-space of $(A - \lambda I_n)$.
- ▶ Every non-zero member of E_λ is an eigen-vector associated with λ .
- ▶ $\mu_A(\lambda)$ = dimension of E_λ = geometric multiplicity of λ .
- ▶ $\nu_A(\lambda)$ = multiplicity of λ as a root of $(p_n(A) = 0)$ = algebraic multiplicity of λ .
- ▶ $\mu_A(\lambda) \leq \nu_A(\lambda)$ for every A and λ .

Eigenvalues and Eigenvectors

- ▶ $x_1 \in E_{\lambda_1}, x_2 \in E_{\lambda_2}, \lambda_1 \neq \lambda_2 \implies x_1, x_2$ are independent.
- ▶ $x_1, \dots, x_k$ are eigenvectors associated with *distinct* eigenvalues $\lambda_1, \dots, \lambda_k \implies \{x_1, \dots, x_k\}$ is an independent set.
- ▶ **Corollary** : $\lambda_1, \dots, \lambda_k$ are distinct. B_i is a basis for E_{λ_i} for $i = 1, \dots, k$. Then, $B_1 \cup \dots \cup B_k$ is an independent set.
- ▶ A basis $B = \{x_1, \dots, x_n\}$ of R^n is an *eigenbasis* w.r.t. $A \in R^{n \times n}$, if each x_i is an eigenvector of A .
- ▶
- ▶ **Defn** : A and B are *similar* if, for some invertible P , we have $A = PBP^{-1}$.
- ▶ A and B are similar if they represent the same LT but for possibly different bases of R^n . It is an equivalence relation over $\{A : A \in R^{n \times n}\}$.
- ▶ **Defn** : A is *diagonalisable* if A is similar to some diagonal $D = \text{diag}(a_1, \dots, a_n)$.
- ▶ If A is diagonalisable, then $\det(A) = \det(D) = \prod_{i=1}^n a_i$.
- ▶ If A is diagonalisable, then $A^k = PD^kP^{-1}$, for some invertible P , leading to faster computation of A^k .
- ▶ If $t(n)$ denotes the asymptotic number of flops of best algorithm for computing the product of two $n \times n$ matrices over reals, then a straight-forward computation requires $O((\log k)t_n)$ flops, whereas, for diagonalisable A , it can be done in $O(2t_n + n(\log k))$ flops.
- ▶ **Theorem** : $A = PDP^{-1}$ for some $D = \text{diag}(a_1, \dots, a_n) \in R^{n \times n} \iff a_i$'s are eigen values of A and columns of $P = [p_1, \dots, p_n]$ are the corresponding eigenvectors of A .
- ▶ **Proof** : $A = PDP^{-1} \iff AP = PD \iff Ap_i = a_i p_i$ for each $i \iff a_i$'s are eigenvalues and $\{p_1, \dots, p_n\}$ is an eigenbasis of A .

Eigenvalues and Eigenvectors

- ▶ An *orthonormal eigenbasis* (ONEB) w.r.t. $A \in R^{n \times n}$, is an orthonormal basis $B = \{x_1, \dots, x_n\}$ which is also an eigenbasis w.r.t. A .
- ▶
- ▶ **Spectral Theorem** : For a symmetric $A \in R^{n \times n}$, all of its eigenvalues are reals. Also, A admits an orthonormal eigenbasis (ONEB) B for R^n .
- ▶ **Corollary** : Let $A \in R^{n \times n}$. Then, A is symmetric $\iff$ there exists an ONEB w.r.t A for $R^n \iff \exists$ an orthogonal $U \in R^{n \times n}$ and a diagonal $D = \text{diag}(a_1, \dots, a_n) \in R^{n \times n}$ such that $A = UDU^T$.
- ▶ **Proof of Corollary** : (1) $\implies$ (2) follows from Spectral Theorem. For (2) $\implies$ (3), suppose that there are $\lambda_1, \dots, \lambda_n \in R$ and an ONEB $B = [x_1, \dots, x_n]$ w.r.t. A such that $Ax_i = \lambda_i x_i$ for each i . We have $AB = [Ax_1, \dots, Ax_n] = [\lambda_1 x_1, \dots, \lambda_n x_n] = BD$, where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. This implies that $A = ABB^T = BDB^T$ proving the existence of required matrices. For (3) $\implies$ (1), assume that $A = UDU^T$ for orthogonal B and diagonal D . Then, $A^T = (UDU^T)^T = UDU^T = A$, proving that A is symmetric.
- ▶ **Geometric interpretation** : A and D represent the same LT $T : R^n \rightarrow R^n$ but with respect to different bases. That is, A represents T w.r.t. standard basis $\{e_1, \dots, e_n\}$ and D represents T w.r.t. ONEB formed by the columns of U . U represents a rotational transformation, that is, norms and distances remain unchanged.

Eigenvalues and Eigenvectors

► **Corollary :** Let $A \in \mathbb{R}^{n \times n}$ be symmetric. Then,

- (1) A is positive semi-definite (positive definite) $\iff$ all eigenvalues of A are nonnegative (are positive).
(2) A is positive semi-definite (positive definite) $\iff A = BB^T$ for some $B \in \mathbb{R}^{n \times n}$ (for some invertible $B \in \mathbb{R}^{n \times n}$).

►

► **Proof of (1) :** $\implies$: Let λ_i be any eigenvalue with associated x_i . Then,
 $x_i^T A x_i = \lambda_i x_i^T x_i \geq 0 \Rightarrow \lambda_i \geq 0$. $\impliedby$: Assume w.l.o.g. that $B = [x_1, \dots, x_n]$ forms an ONEB. Then,
for any $x = \sum_i \alpha_i x_i$, we have $x^T A x = \sum_i \alpha_i^2 \lambda_i \geq 0$ and hence A is PSD. Similar arguments prove the case of positive definite matrices.

► **Proof of (2) :** $\implies$: Write $A = UDU^T$ where U is orthogonal and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_i \geq 0$ for each i . Write $B = UD^{1/2}$. Then $B^T = D^{1/2}U^T$ and hence $A = BB^T$, as required. When A is PD, we have $\lambda_i > 0$ for each i and hence B is invertible. $\impliedby$: $A = BB^T \implies x^T A x = y^T y \geq 0$ with $y = B^T x$, for each x . Hence A is PSD. When B is invertible, y is non-zero for each $x \neq 0$ and hence $x^T A x > 0$ for each $x \neq 0$. This shows that A is PD.

Singular Value Decomposition

- ▶ **SVD Theorem** : For every $A \in R^{m \times n}$ with $\text{rank}(A) = r$, there exist orthogonal matrices $U \in R^{m \times m}$, $V \in R^{n \times n}$ and a matrix $\Sigma \in R^{m \times n}$ satisfying $A = U\Sigma V^T$, where $\Sigma_{ii} = \sigma_i \geq 0$ for $i \leq r$ and $\Sigma_{ij} = 0$ for every other (i, j) . $\sigma_1, \dots, \sigma_r$ are known as A 's singular values.
- ▶ **Geometric interpretation** : A and Σ represent the same LT $T : R^n \rightarrow R^m$ but with respect to different bases. That is, A represents T w.r.t. standard bases $\{e_1, \dots, e_n\}$ and $\{e_1, \dots, e_m\}$ respectively for R^n and R^m . Σ represents T w.r.t. orthonormal bases for R^n and R^m , formed by the columns of V and U respectively.
- ▶ **Geometric Interpretation** : (1) Apply the rotation specified by V^T to change from standard basis of R^n to the basis formed by the columns of V ; (2) map each v_i ($i \leq r$) to $\sigma_i u_i$ and map each v_i ($i > r$) to 0; (3) Apply the rotation specified by U to change from basis formed by the columns of U to the standard basis of R^m .
- ▶ **Corollary** : Writing $U = [u_1, \dots, u_m]$ and $V = [v_1, \dots, v_n]$, we have $Av_i = U\Sigma e_i = \sigma_i u_i$ for each $i = 1, \dots, r$. Also, $Av_i = U\Sigma e_i = 0$ for each $i > r$.
- ▶ **Corollary** : $A = [\sigma_1 u_1, \dots, \sigma_r u_r, 0, \dots, 0]V^T = \sum_{i=1}^r \sigma_i u_i v_i^T$. Hence, A is the sum of r rank-1 $m \times n$ matrices.
- ▶ **Defn** : The *spectral norm* of a matrix $A \in R^{m \times n}$ is defined as $\max_{x \in R^n} \frac{\|Ax\|_2}{\|x\|_2}$.
- ▶ **Corollary** : A , U , V and Σ be as before. Then, $\|A\|_2 := \max_{x \in R^n} \frac{\|Ax\|_2}{\|x\|_2} = \max_{i \leq r} \sigma_i$.
- ▶ **Proof** : Assume w.l.o.g. that $\sigma_1 = \max_{i \leq r} \sigma_i$. Let $x = \alpha_1 v_1 + \dots + \alpha_n v_n$ be any element of R^n . Then, $Ax = \sum_{i=1}^r \sigma_i u_i v_i^T x = \sum_{i=1}^r \sigma_i u_i \alpha_i$. Hence $\|Ax\|_2^2 = \sum_{i=1}^r \alpha_i^2 \sigma_i^2 \leq \sigma_1^2 \left(\alpha_1^2 + \dots + \alpha_n^2 \right) \leq \sigma_1^2 \|x\|_2^2$. Hence, $\max_x \frac{\|Ax\|_2}{\|x\|_2} \leq \sigma_1$. Also, for $x = v_1$, we have $\frac{\|Ax\|_2}{\|x\|_2} = \sigma_1$, thereby proving the equality.

SVD Theorem and its consequences

- ▶ **Eckart-Young Theorem** : For any $A \in R^{m \times n}$ with $\text{rank}(A) = r$, $A = U\Sigma V^T$ with singular values $\sigma_1 \geq \dots \geq \sigma_r \geq 0$, define $A(k) = \sum_{i=1}^k \sigma_i u_i v_i^T$. It holds that $B = A(k)$ achieves the least value of $\|A - B\|_2$ among all $B \in R^{m \times n}$ with $r(B) \leq k$. That is, $A(k) = \arg\min_{r(B) \leq k} \|A - B\|_2$. Also, $\|A - A(k)\|_2 = \sigma_{k+1}$.
- ▶ **SVD Theorem - Equivalent version** : For every $A \in R^{m \times n}$ with $\text{rank}(A) = r$, there exist ordered orthonormal collections $u_1, \dots, u_r \in R^m$ and $v_1, \dots, v_r \in R^n$ of k vectors each, satisfying $A = \sum_{i=1}^r \sigma_i u_i v_i^T$, where $\sigma_1, \dots, \sigma_r \geq 0$.
- ▶
- ▶ **Defn** : For $A \in R^{m \times n}$ with $\text{rank}(A) = r$ and for $k \leq r$, a *rank- k approximation* of A is a matrix $B \in R^{m \times n}$ with $\text{rank}(B) \leq k$.
- ▶ By EY Theorem, $A(k)$ is an optimal rank- k approximation B of A which minimizes $\|A - B\|_2$.
- ▶ $A(k)$ gives the optimal projection of A onto the subspace formed by lower ranked (i.e. $\text{rank} \leq k$) matrices.

Computation of SVD

- ▶ Given $A \in R^{m \times n}$ with $r = \text{rank}(A)$, construct a SVD of A .
- ▶
- ▶ **Computation :**
- ▶ Compute $V = [v_1, \dots, v_n] \in R^{n \times n}$, $D = \text{Diag}(\lambda_1, \dots, \lambda_n)$ satisfying $A^T A = V D V^T$, where
- ▶ λ_j 's are nonnegative eigenvalues of $A^T A$ and $\lambda_j = 0$ for $j > r$ and
- ▶ $[v_1, \dots, v_n]$ forming an ONEB of $A^T A$.
- ▶ Compute $\sigma_j = \sqrt{\lambda_j}$ for each $j \leq r$.
- ▶ Compute Σ defined by $\Sigma_{ij} = \sigma_i$ if $i = j \leq r$ and $\Sigma_{ij} = 0$ otherwise.
- ▶ Compute $u_i = (A v_i) / \sigma_i$ for each $i \leq r$.
- ▶ $\{u_1, \dots, u_r\}$ is an orthonormal collection. Extend it to an ONB $U := [u_1, \dots, u_m]$ of R^m .
- ▶ Output (U, Σ, V) as the SVD of A .